Errata and minor updates

Physical Geodesy: A Theoretical Introduction

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1. Page 3, lines 8 to 9:
   Change "as attraction, of m" to "as attraction of m"

2. Page 5. Fig. 1.3 caption:
   Change "acceleration" to "attraction"

3. Page 16, lines 2,3,5, 6 and 7:
   Change "$\hat{e}_h$" to "$\hat{e}_H$"

4. Page 16, line 7:
   Change "$\partial W/\partial h$" to "$\partial W/\partial H$"

5. Page 16, eqn (1.48):

$$g = -\frac{\partial W}{\partial H}\,.$$

6. Page 16, eqn (1.49):

$$dH = -\frac{dW}{g}\,. \tag{1}$$

7. Page 26. Below eqn (1.85) and below eqn (1.88):
   Change "disc" to "disk"

8. Page 37. Line 7 from bottom:
   Change "A function satisfying" to 'A function with continuous second order partial derivatives satisfying"

9. Page 40, below eqn. (1.151):
   $(\Delta x^2 + \Delta z^2)^{1/2} = [1 + (dz/dx)^2]^{1/2}\Delta x$

10. Page 44. Line 11:
Change "Fig. 1.17b." to "Fig. 1.17b, "where is curve is an infinitesimal fraction."

11. Page 61, add the following paragraph after the last line:
Readers are suggested to study the uniqueness of the solutions of the boundary value problems for an internal domain, and furthermore, to show that the surface density μ in (2.55) is unique based on the uniqueness of the solutions of the first-type boundary value problems for both an external domain and an internal domain (suggestion: assume V is known on S, and prove that C does not contribute to $(\partial U/\partial n)_i$). Readers are also suggested to compare the expression of μ in (2.55) with (1.130) by setting $\hat{e}_t = \hat{e}_n$.

12. Page 89. eqn (3.45):

$$\delta = \lim_{i\to\infty}\left|\frac{C_i}{C_{i+2}}\right| = \lim_{i\to\infty}\left|\frac{(i+2)(i+1)}{i(i+1)-\alpha}\right| = 1\,.$$

13. Page 91, eqn (3.58):

$$y_1^0(x) = \sum_{i=1}^{\infty}\frac{1}{2i}x^{2i} \qquad \text{and} \qquad y_2^0(x) = \sum_{i=0}^{\infty}\frac{1}{2i+1}x^{2i+1}$$

14. Page 91, add to the end of the paragraph after (3.58):
The factor M in (3.57) is dropped, since it does not affect the convergence.

15. Page 91, eqn (3.60):

$$\begin{cases} \dfrac{1}{2}[\ln(1+x)+\ln(1-x)] = -\displaystyle\sum_{i=1}^{\infty}\frac{x^{2i}}{2i}\,, \\ \dfrac{1}{2}[\ln(1+x)-\ln(1-x)] = \displaystyle\sum_{i=0}^{\infty}\frac{x^{2i+1}}{2i+1}\,. \end{cases}$$

16. Page 105, eqn (3.132):

$$P_n^k(\pm 1) = 0\,, \qquad k \neq 0\,;$$

17. Page 105, eqn (3.134):
Replace the last "." by ";"

18. Page 131, eqn (3.270):
Replace "k=1" by "k=0" under the summation sign.

19. Page 136, in equation (3.285):
Change "C_1^0" to "C_1"

20. Page 142, eqn (3.314):

$$V = \frac{GM}{r} \sum_{n=0}^{\infty} \left(\frac{a}{r}\right)^n \sum_{k=0}^{n} (\bar{C}_n^k \cos k\lambda + \bar{S}_n^k \sin k\lambda) \bar{P}_n^k(\cos\theta)\,,$$

Remark: Including the degree 1 normalized coefficients is a better choice.

21. Page 144, eqn (3.321):
Replace “k=1” by “k=0” under the summation sign.

22. Page 144, eqn (3.322):
Replace “k=1” by “k=0” under the summation sign.

23. Page 145, eqn (3.327 should be):

$$\sum_{k=0}^{n} \bar{P}_n^k(\cos\theta) \frac{d}{d\theta} \bar{P}_n^k(\cos\theta) = 0\,.$$

24. Page 153, below eqn (3.370):
Change “Here we ... be written as” to “On the equator, with $\theta = \pi/2$, this formula can be written as”

25. Page 159, last sentence of the Abstract:
Change “called Geodetic Reference System (GRS)” to “which are essential for a Geodetic Reference System (GRS)”

26. Page 188, paragraph below eqn (4.135):
Change “Borkowski (1987)” to “Borkowski (1989)”

27. Page 199, line 7 from bottom:
Change “which are referred to as Geodetic Reference System” to “which are essential for a Geodetic Reference System”

28. Page 200, the line above eqn (4.180):
Change “are the Geodetic Reference System GRS80” to “are those of the Geodetic Reference System 1980 (GRS80)”

29. Page 201. End of the line above eqn (4.195):
Add “,”

30. Page 202, line 7 from bottom, the reference should be replaced by:
Borkowski, K.M. (1989). Accurate algorithms to transform geocentric to geodetic coordinates. Bulletin Géodésique, 63, 50-56.

31. Page 204, the first line of the paragraph above the Section title 5.1.2:
Change “defers” to “differs”

32. Page 209, eqn (5.20):

$$\left(\frac{\partial U}{\partial n_\Sigma}\right)_\Sigma = \left(\frac{\partial U}{\partial n_S}\right)_S + N \left(\frac{\partial^2 U}{\partial n_S^2}\right)_S = \left(\frac{\partial U}{\partial n_S}\right)_S - N \left(\frac{\partial \gamma}{\partial n_S}\right)_S\,.$$

33. Page 246, Title of Sect. 5.5.2:
Change “Degree Power Spectrum” to “Degree Variance Power Spectrum”

34. Page 247, eqn (5.172):

$$V_n = \frac{GM}{R}\left(\frac{R}{r}\right)^{n+1}\sum_{k=0}^{n}\left[\frac{2(2n+1)}{1+\delta_k}\frac{(n-k)!}{(n+k)!}\right]^{1/2}$$
$$\times(\bar{C}_n^k\cos k\lambda+\bar{S}_n^k\sin k\lambda)P_n^k(\cos\theta)\,,$$

35. Page 247, eqn (5.174):

$$\frac{1}{4\pi}\int_\omega V^2 d\omega = \left(\frac{GM}{R}\right)^2\sum_{n=0}^{n_{\max}}\left(\frac{R}{r}\right)^{2n+2}\sum_{k=0}^{n}\left[\left(\bar{C}_n^k\right)^2+\left(\bar{S}_n^k\right)^2\right]$$
$$= \sum_{n=0}^{n_{\max}}\left(\frac{1}{4\pi}\int_\omega (V_n)^2 d\omega\right).$$

36. Page 248, Title of Fig. 5.10:
Change “Degree power spectrum variance” to “Degree variance power spectrum”

37. Page 249, Title of Sect. 5.5.3:
Change “Degree Power Spectrum” to “Degree Variance Power Spectrum”

38. Page 256, line 4 from bottom:
Delete the sentence “The correction $\Delta_1 g + \Delta_2 g$ is referred to as incomplete Bouguer correction”

39. Page 256, Last line:
Change “Bouguer correction and anomaly” to “Bouguer anomaly”

40. Page 257, lines 2-3 from bottom:
Change “Evidently, $l^2 = r^2 + z^2$, $\sin\beta = z/(r^2+z^2)^{1/2}$ and $dv = r d\alpha dr dz$” to “Evidently, $\sin\beta = z/l$, $l = (r^2+z^2)^{1/2}$ and $dv = r d\alpha dr dz$.”

41. Page 258, eqn (6.13):

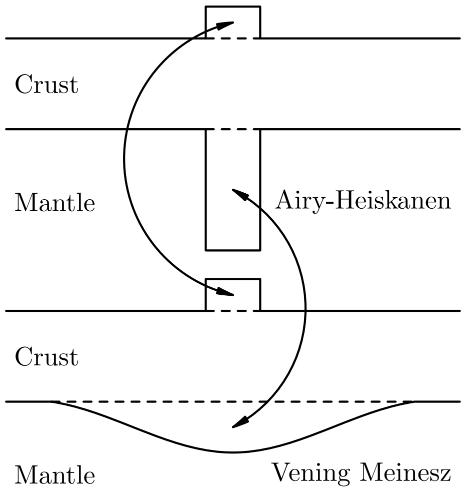
$$\Delta_3 g = G\rho\int_0^{2\pi} d\alpha\int_0^a dr\int_0^{\Delta H}\frac{zr}{(r^2+z^2)^{3/2}}dz\,,$$

42. Page 258:
Change the sentence above (6.15) to: “The gravity anomaly computed according to”

43. Page 263, eqn (6.27):

$$\Delta\rho = \rho - \rho' = -\frac{(\rho-\rho_w)P}{D-P}\,.$$

44. Page 265, Fig. 6.7 (It makes me feel embarrassed to have the one in the print! I went to replace it before the submission of the final version to Springer, but I have mistakenly replaced it in a different draft):



45. Page 301, Line 5:
Change "$\sim$ 10 m for the isostatic anomaly." to "$\sim$ 10 m for the isostatic anomaly (Heiskanen and Moritz, 1967)."

46. Page 313, eqn (6.223):

$$f_T(T, H', \psi) = \frac{R^2}{2}\Bigg\{(x + 3\cos\psi)(1 + x^2 - 2x\cos\psi)^{1/2} + (3\cos^2\psi - 1) \times \ln\left[(1 + x^2 - 2x\cos\psi)^{1/2} + x - \cos\psi\right]\Bigg\}_{1-(T+H')/R}^{1-T/R}.$$

47. Page 319:
Add the following into the references:
Heiskanen, W.A., & Moritz,H. (1967). *Physical geodesy*. W.H. Freeman and Company.

48. Page 319, add to further reading:
Szwillus, W., Ebbing, J., & Holzrichter, N. (2016). Importance of far-field topographic and isostatic corrections for regional density modelling. *Geophysical Journal International, 207*, 274–287. https://doi.org/10.1093/gji/ggw270

49. Page 393 line 3 and page 493 line 7 from bottom (So embarrassed to misspell a colleague's name):
Change "Bgherbandi" to "Bagherbandi".

The same correction applies to Page 493, line 7 from the bottom.

50. Page 398, line 26 (or line 9 of the third paragraph):
    Change "If not, the difference between the differences" to "If not, the sum of the two differences"

51. Page 410, line 6:
    Change "$n_{\max}(n_{\max}+1)$" to "$(n_{\max}+1)^2$"

52. Page 415, line 2 of the last paragraph:

    Add ", although the decrease may be oscillatory" after "β decreases with n"

53. Page 417, line 3 from bottom:
    Change "$\int_{-\sigma}^{\sigma} g(x)dx = 99.73.$" to "$\int_{-3\sigma}^{3\sigma} g(x)dx = 99.73\%.$"

54. Page 419:
    Add a "," after (8.75) just above eqn (8.78).

55. Page 423: Eqn (8.98) should be

$$\sum_{i=1}^{N}\sum_{j=1}^{2N} W_{ij}\left[\sum_{n=2}^{N-1}\sum_{k=0}^{n}(\alpha_{ijnk}\bar{c}_n^k + \beta_{ijnk}\bar{s}_n^k) - \Delta\bar{g}_{ij}\right]^2 = \min$$

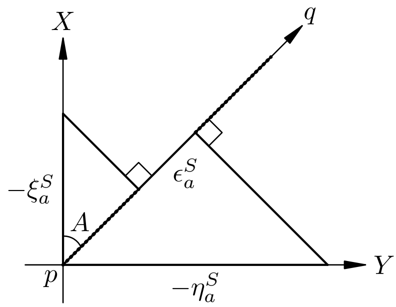
56. Page 424: Change the paragraph starting the 3rd line to
    "The second problem is the resolvability. In this approach, the coefficients $\bar{c}_n^k$ and $\bar{s}_n^k$ can be reliably solved only up to degree and order $N-1$ (Colombo, 1981). This could be demonstrated by examining the problem on the equator where $\theta = \pi/2$ and (8.32) degenerates to the Fourier series

$$\Delta g(\lambda) = \sum_{k=0}^{n_{\max}}(a_k \cos k\lambda + b_k \sin k\lambda)\,. \tag{8.99}$$

    Hence, there are $2n_{\max}+1$ coefficients to be determined. However, there are $2N$ values of Δg on the equator. The total number of coefficients to be determined cannot be more than the total number of data. As a result, the coefficients can only be solved up to order $n_{\max} = N-1$ at most. With $n_{\max} = N-1$ as chosen in (8.98), the total number of coefficients $\bar{c}_n^k$ and $\bar{s}_n^k$ is N^2, while the total number of data $\Delta\bar{g}_{ij}$ is $2N^2$."

57. Page 438, second line below 8.5.1:
    Change "base" to "based"

58. Page 439, Fig. 8.13:



59. Page 450. eqn (9.14):
    Change "$\frac{d}{\partial r}$" to "$\frac{d}{dr}$"

60. Page 452. eqn (9.19):
    Change "$Y_n^k(\theta, \lambda)$" to "$Y_n(\theta, \lambda)$"

61. Page 461, the line above (9.66):
    Change "We now assume" to "We now make the finer approximations"

62. Page 461, the line below (9.66):
    Change "this assumption" to "these approximations"

63. Page 487. Add the following by the beginning of the line below eqn (A.57):
    The substitution of (A.54) into this equation yields an equation for B, which is referred to as the latitude equation in the literature.